

# Advanced Natural Language Processing

## Discussion 11: Speech and Multi-Modality

Department of Advanced Computer Sciences (DACS)  
Master of AI / DSDM  
Prof dr ir Jan C. Scholtes – Course year 2026-2027



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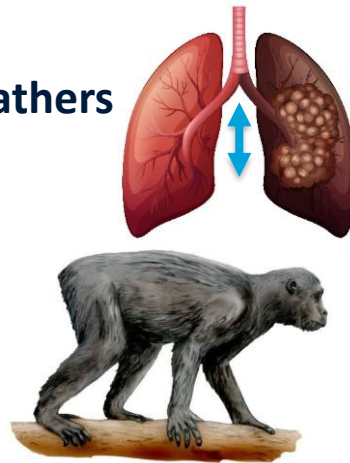
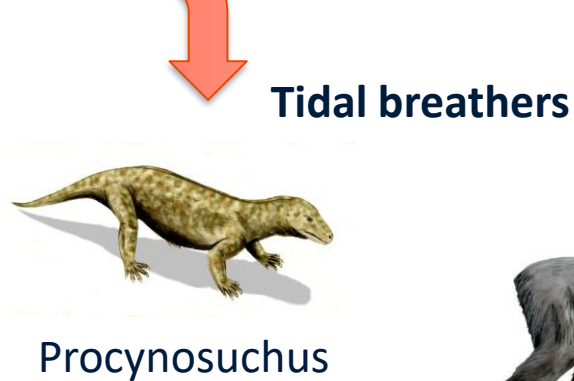
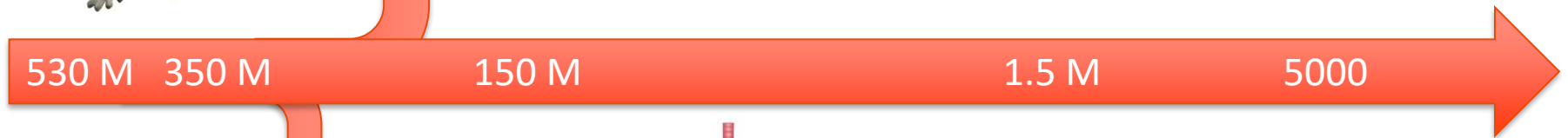
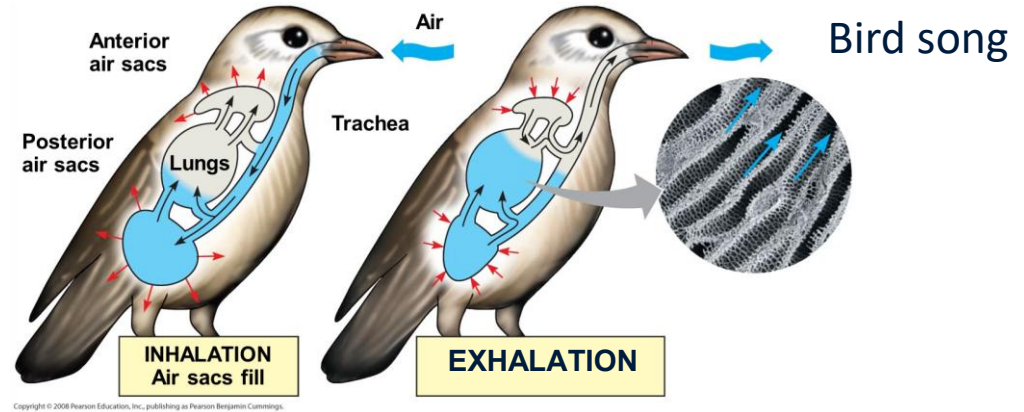
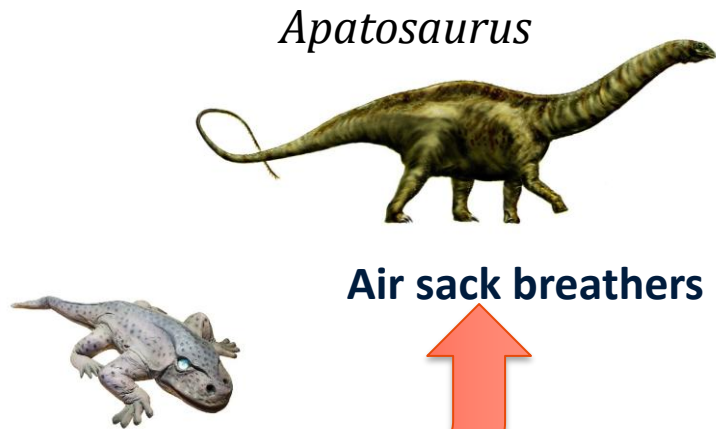


# Speech Recognition



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# Vocal communication – evolutionary foundations



Mammals,  
vocal communication



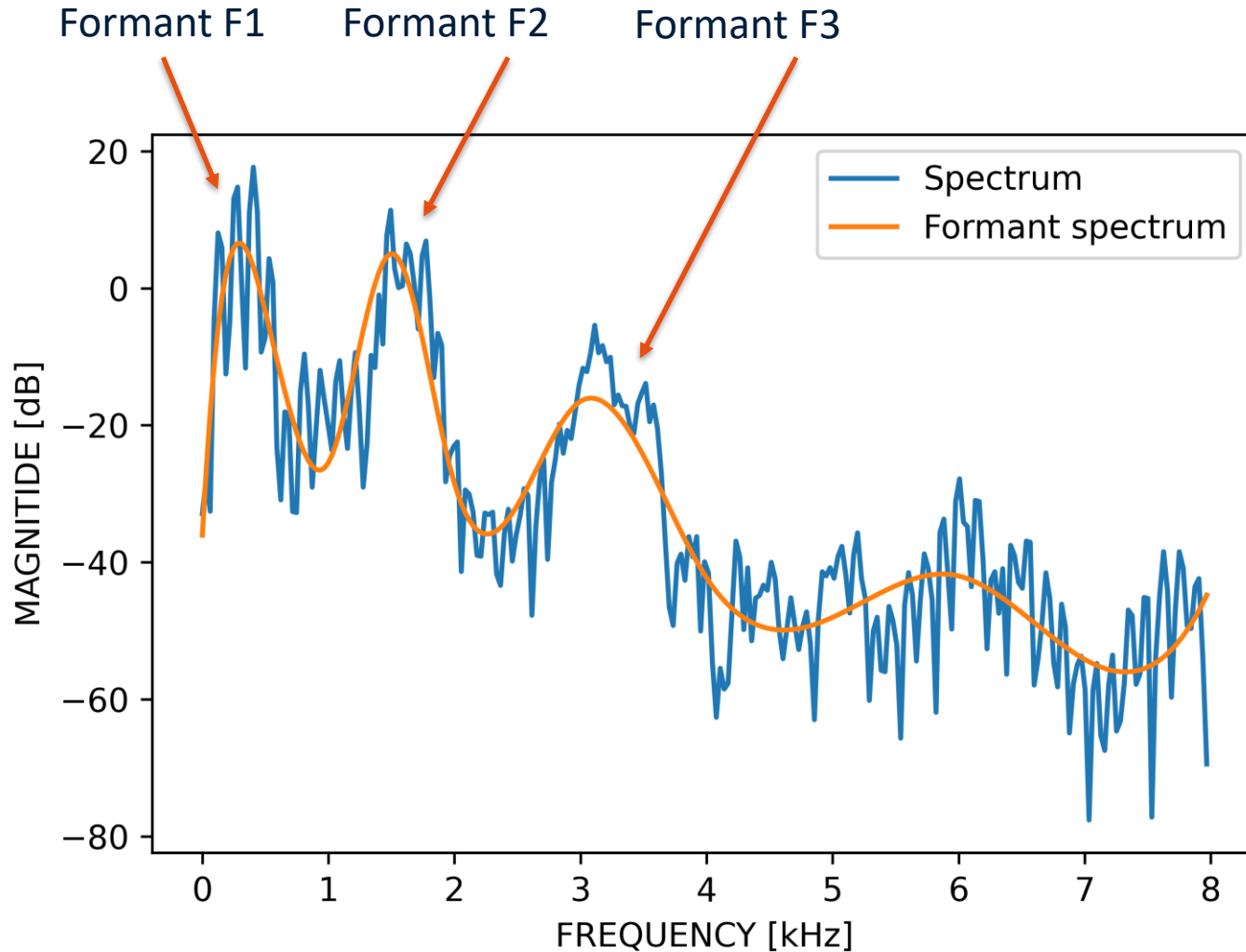
**Homo erectus (1.5M) – spoken language**

- Written language (3000 BC)
- Printing (1450 AD)
- Speech transmission (1876)
- Computer speech (1950)
- Large Language Models (2018)

# Spectrum formants

Formants – vocal tract

Fine-structure - the excitation



# Human Speech: Phonemes/Phones

- **Phones:**
  - Basic speech sounds (around 200-300)
- **Phonemes:**
  - Defined as fundamental units of speech
  - If you change it, it (can) change the meaning
  - Language dependent
- **Vowels – voiced speech**
  - Voiced speech, glottis exciting open vocal tract
  - There are six vowels in the English language: a, e, i, o, u and sometimes y.
- **Consonants – unvoiced speech:**
  - sound segments in which the airflow is at least partly obstructed
  - The 21 consonant letters in the English alphabet are B, C, D, F, G, H, J, K, L, M, N, P, Q, R, S, T, V, X, Z, W and Y.



# Classic ASR systems (1980's-to-2010's)



Fast Fourier Transform (FFT) - speech to spectrum

Mel-spectrum mapping (and Mel-cepstrum by another FFT) - to auditory representation, Decibel magnitude (logarithm), or cepstrum

TRAINED WITH LARGE COLLECTION OF TRANSCRIBED SPEECH DATA

Model spectra by Gaussian Mixture Models (GMMs) – model spectrum “states”

Hidden Markov Model (HMM) – model state transitions by a probabilistic model

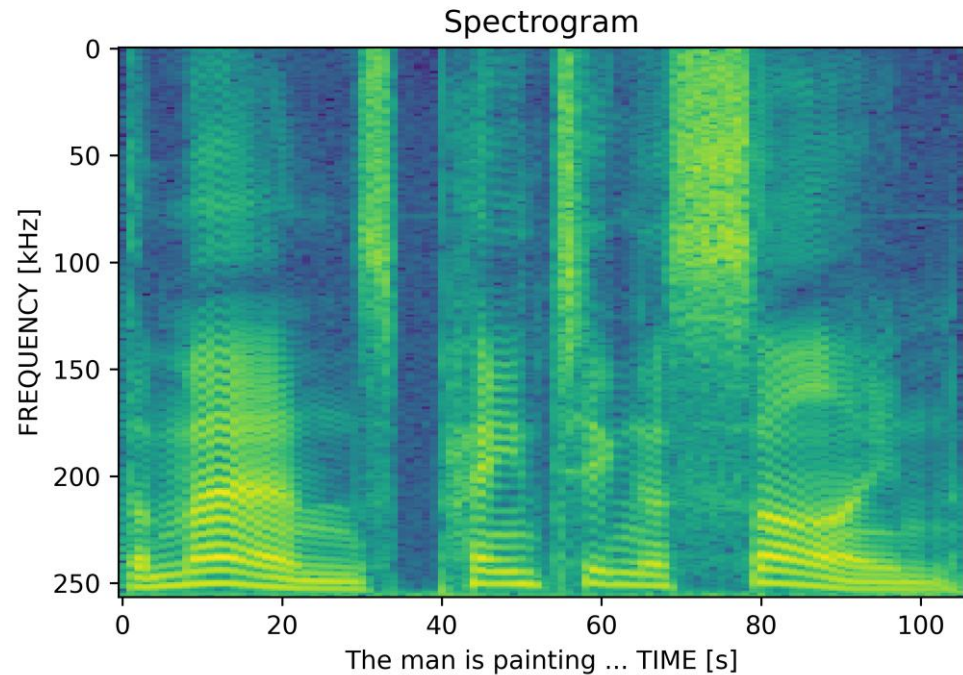
Decoding of the most likely phoneme sequence by Viterbi algorithm

“The man is painting a sign”

# A Spectrogram

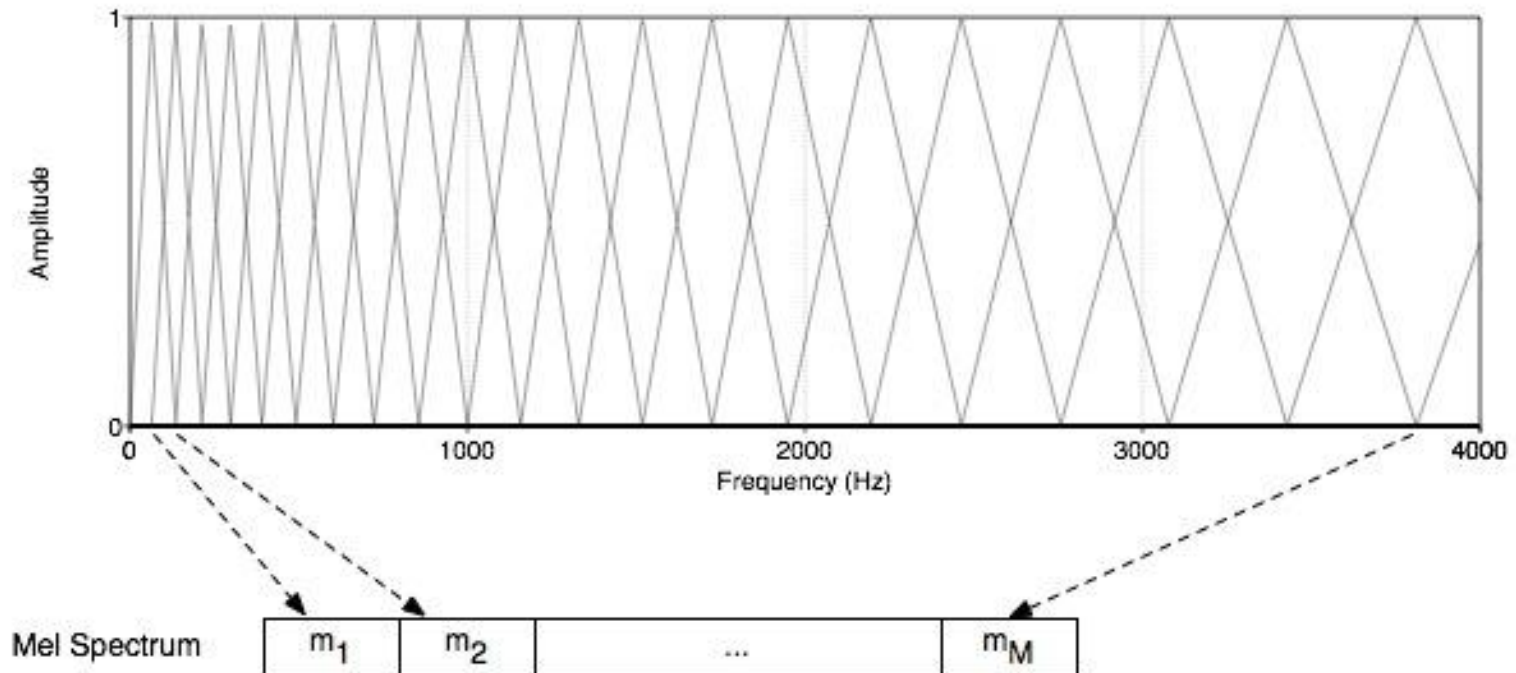
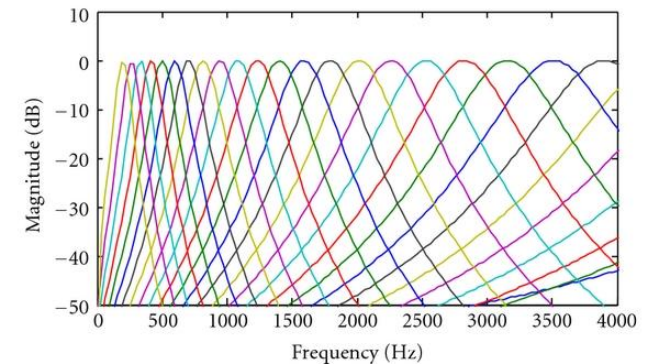
- A spectrogram represents a number of FFTs stacked on top of each other.
- It is a way to visually represent a signal's loudness, or amplitude, as it varies over time at different frequencies.
- The y-axis is converted to a log scale, and the color dimension is converted to decibels (e.g. the log scale of the amplitude).

Intuition: humans can only perceive a very small and concentrated range of frequencies and amplitudes.



# Mel Filter Bank Processing

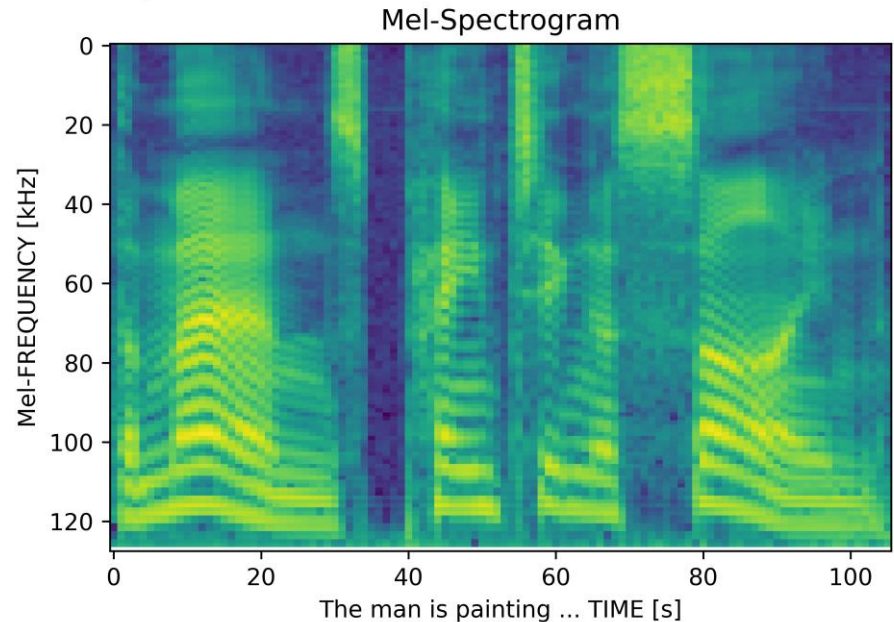
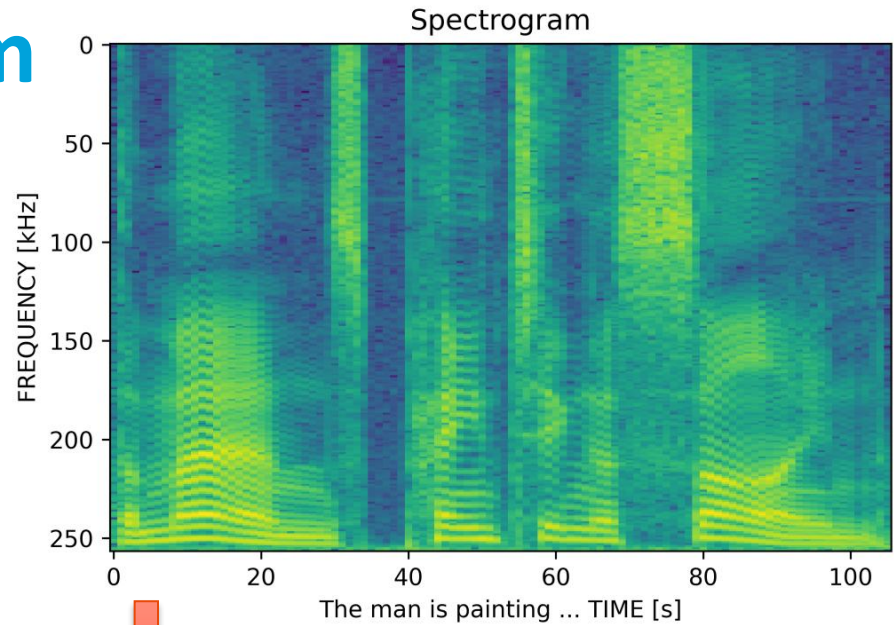
- Mel Filter bank
  - Uniformly spaced before 1 kHz
  - logarithmic scale after 1 kHz





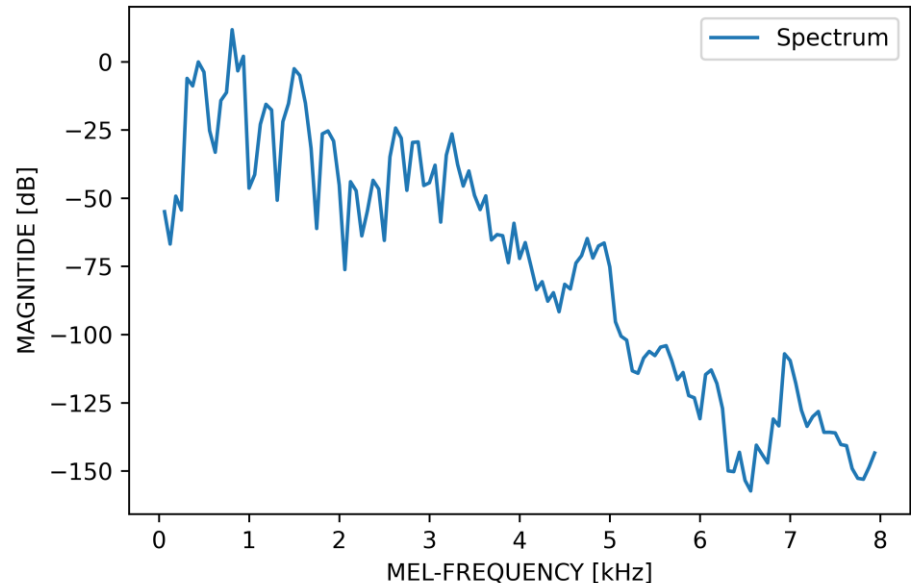
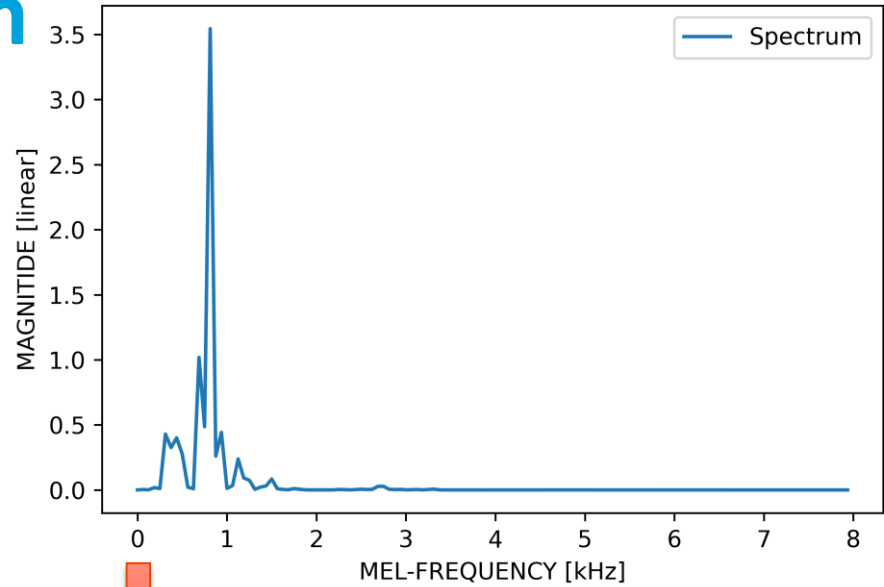
# Log Mel Spectrogram

- A Mel spectrogram is a spectrogram where the frequencies are converted to the Mel scale.
- Spectrum magnitudes are mapped to a logarithmic Decibel scale
- Aka the **Log Mel Spectrogram**
- Matches with:
  - Frequency resolution and loudness perception of hearing
  - Harmonic spectrum of speech production

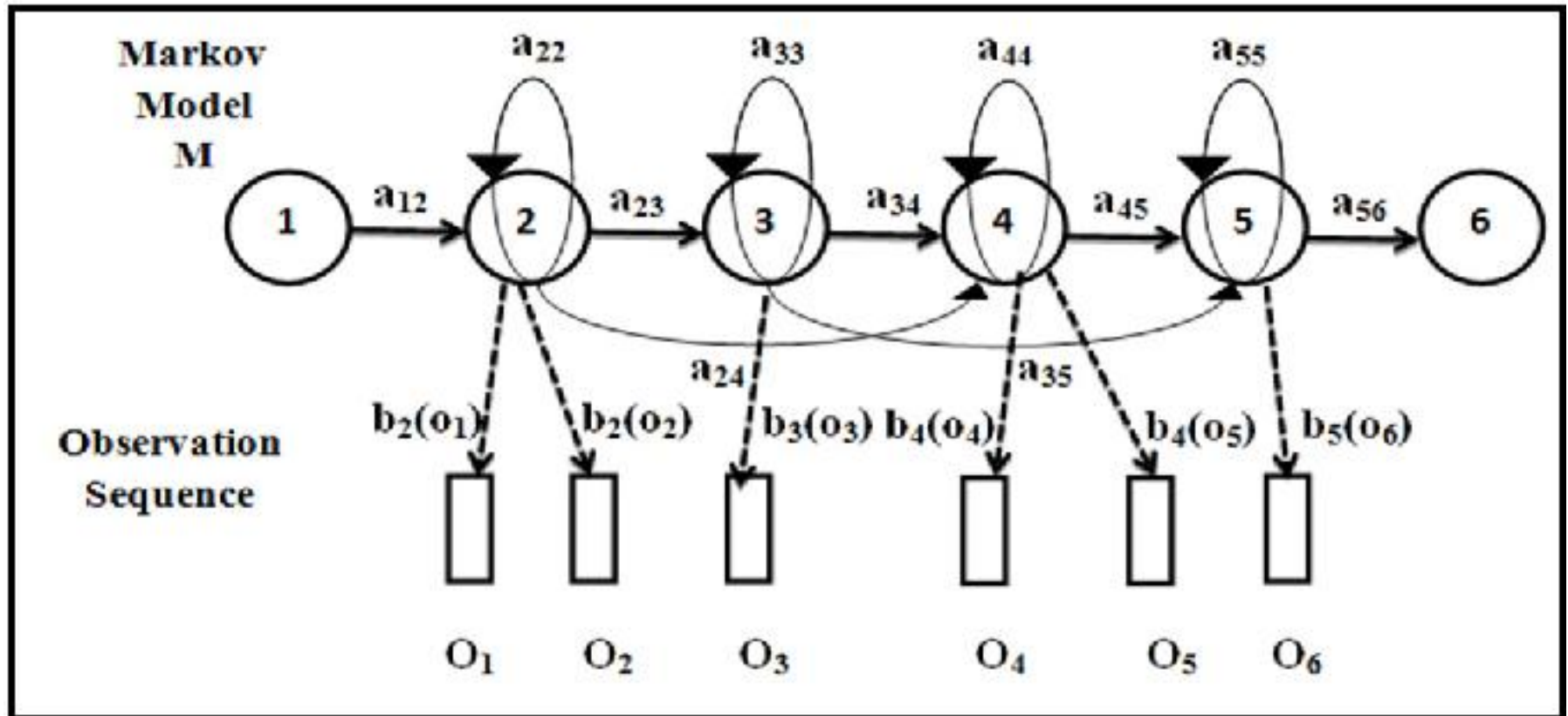


# Log Mel Spectrogram

- Our perception of magnitude of a stimulus follows a logarithmic scale (*Fechner's law, 1860*)
- **Spectrum magnitudes are mapped to a logarithmic Decibel scale**
- Aka the **Log Mel Spectrogram**
- Matches with:
  - Frequency resolution and loudness perception of hearing
  - Harmonic spectrum of speech production



# Hidden Markov Model (HMM)



The observations  $O_1 \dots$  can be seen as “tokens” – a learned dictionary of LogMelSpec blobs

The “Hidden Model” has less states than #tokens – one hidden stat may emit multiple tokens



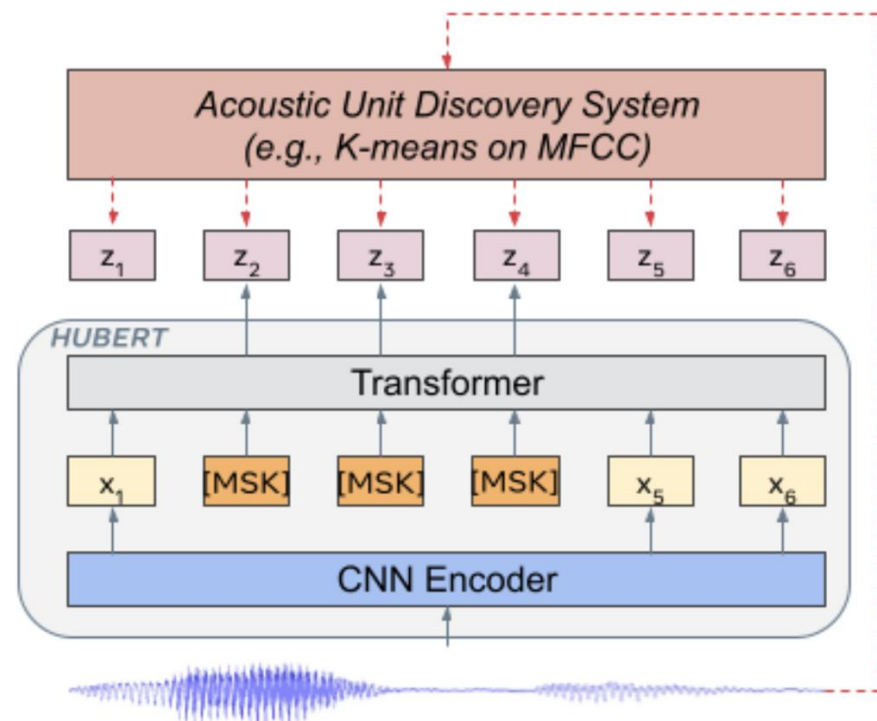
# HuBert speech encoder – like BERT for speech

Spectrogram representation of audio

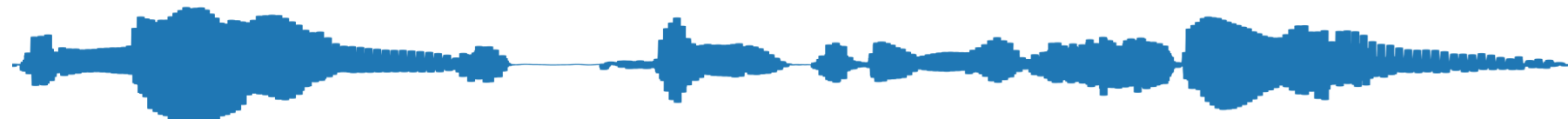
HuBert (a speech model)

- Mel-spectrogram
- CNN encoder to vectors
- Transformer on vectors

Trained in a self-supervised task



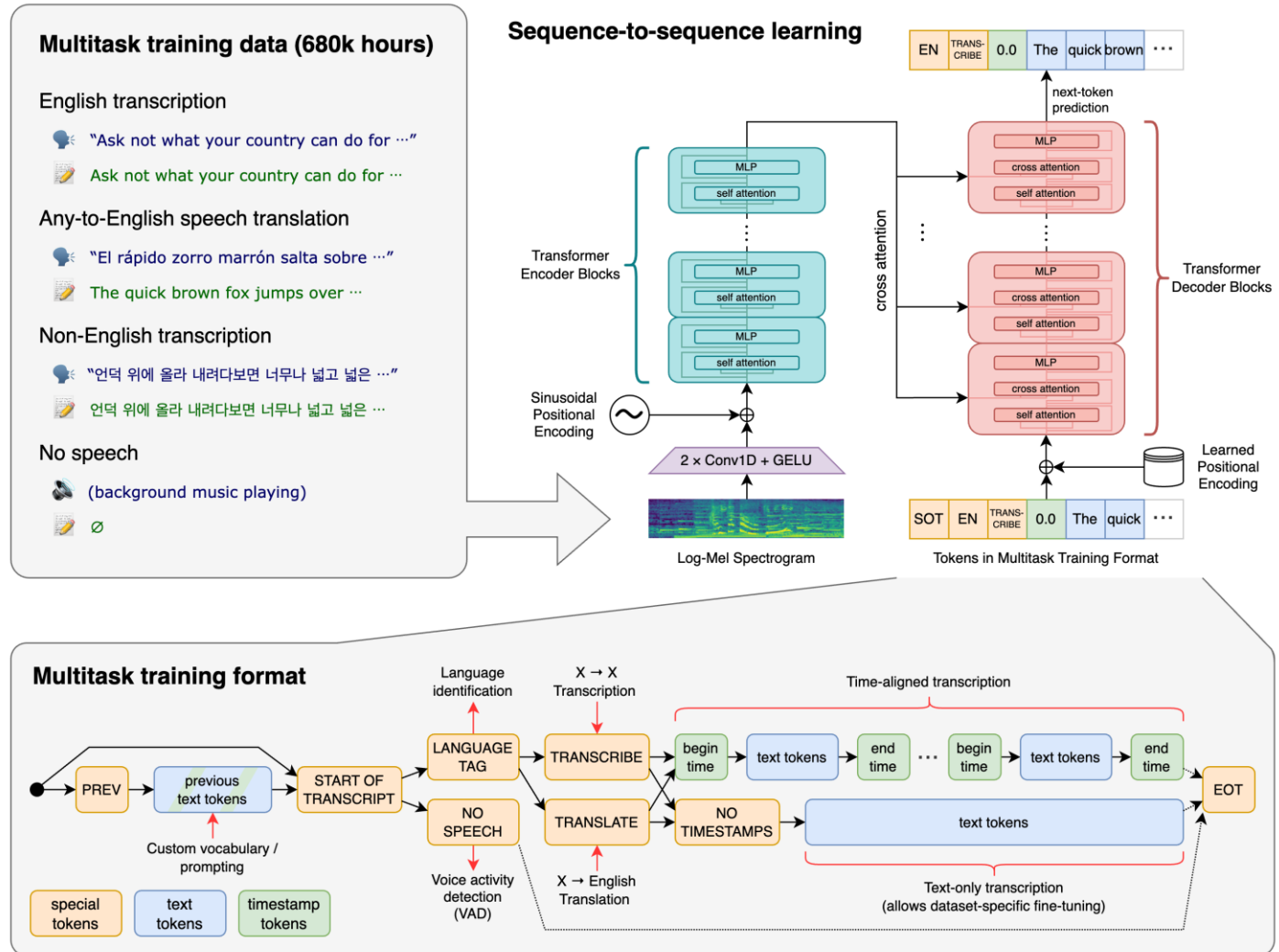
$|q_{79}|q_{688}|q_{584}|q_{386}|q_{335}|q_{810}|q_{603}|q_{148}|q_{452}|q_{243}|q_{186}|q_{630}|q_{807}|q_{222}|q_{38}|q_{614}|q_{442}|q_{862}|q_{364}$



[\[2106.07447\] HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units](#)

# OpenAI: Whisper

## OUR TUTORIAL 9

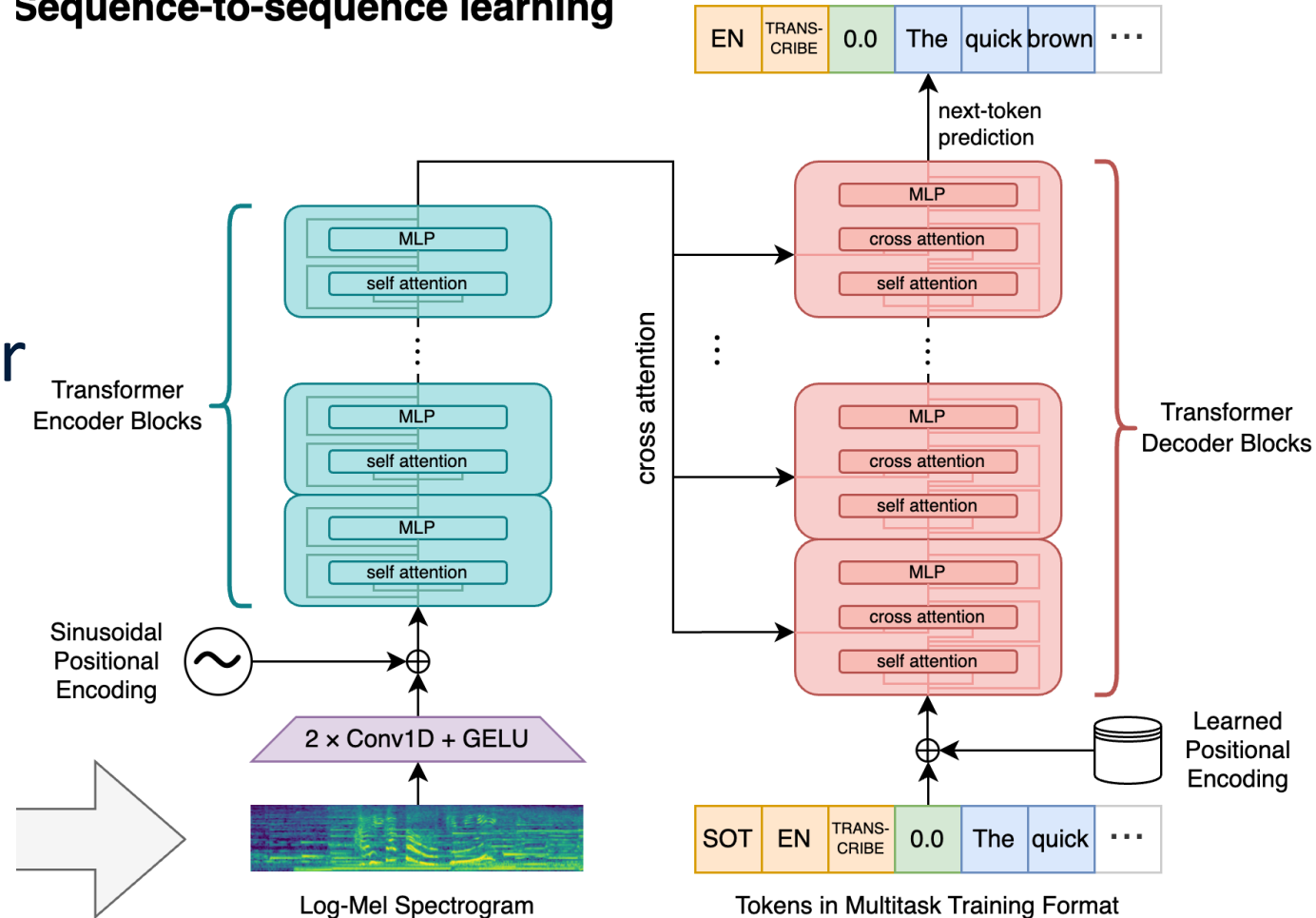


# OpenAI: Whisper

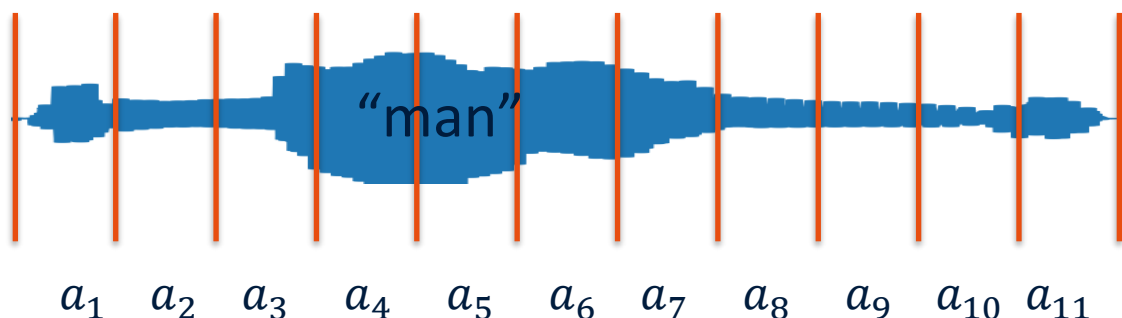
OUR TUTORIAL 9

- Encoder-decoder system
- Transformer

## Sequence-to-sequence learning



# Connectionist Temporal Classification (CTC) – introducing “nothing” $\emptyset$ symbol



m m a a a a a n n n

m  $\emptyset$   $\emptyset$  a  $\emptyset$   $\emptyset$   $\emptyset$   $\emptyset$  n  $\emptyset$   $\emptyset$

Any variant of “man” with 11 symbols  
where  $\emptyset$  is *blank*

m $\emptyset\emptyset\emptyset\emptyset$ a $\emptyset\emptyset\emptyset$ n  
 $\emptyset\emptyset\emptyset\emptyset$ ma $\emptyset\emptyset$ n $\emptyset\emptyset$   
 $\emptyset\emptyset$ ma $\emptyset$ n $\emptyset\emptyset\emptyset\emptyset$   
 $\emptyset\emptyset\emptyset$ m $\emptyset$ a $\emptyset\emptyset\emptyset$ n $\emptyset$   
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**m $\emptyset\emptyset\emptyset$ a $\emptyset\emptyset$ n $\emptyset\emptyset\emptyset$**   
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 $\emptyset\emptyset\emptyset\emptyset$ m $\emptyset\emptyset$ a $\emptyset$ n  
 $\emptyset$ m $\emptyset\emptyset\emptyset\emptyset$ a $\emptyset$ n $\emptyset$

....



# CTC in training



Trans/Conformer



$\mathbf{a} = (a_1 \ a_2 \ a_3 \ a_4 \ a_5 \ a_6)$

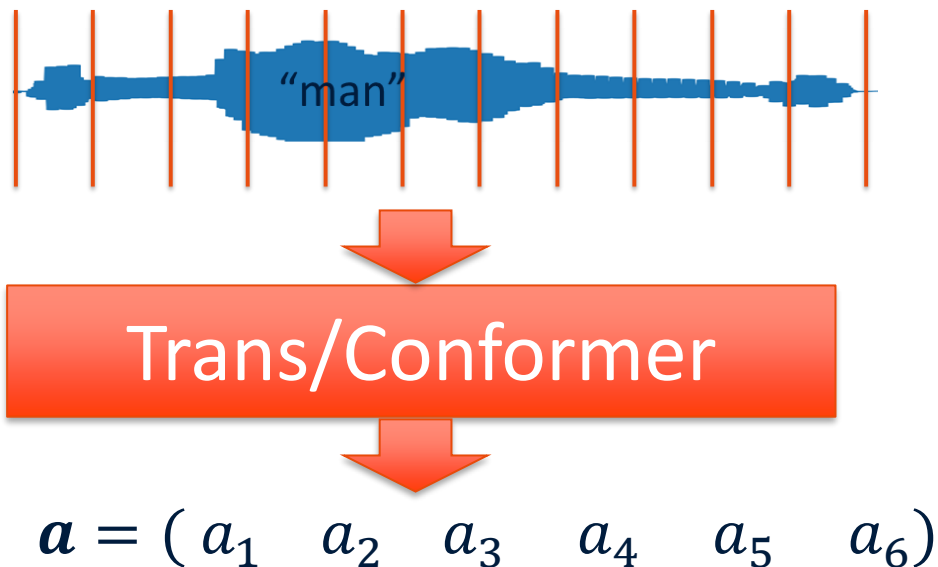
- CTC loss: 
$$e = \frac{1}{N} \sum_i^N L(\mathbf{a}, v_i)$$

man

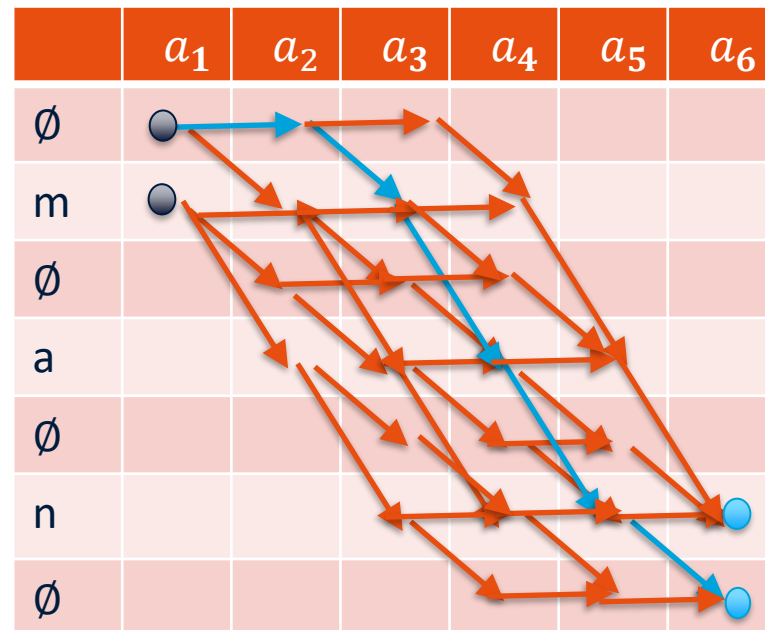
|     |   |   |   |   |   |       |
|-----|---|---|---|---|---|-------|
| m   | a | n | ∅ | ∅ | ∅ | $v_0$ |
| m   | a | ∅ | n | ∅ | ∅ | $v_1$ |
| m   | a | ∅ | ∅ | n | ∅ | $v_2$ |
| m   | a | ∅ | ∅ | ∅ | n | $v_4$ |
| m   | ∅ | a | n | ∅ | ∅ | $v_5$ |
| m   | ∅ | ∅ | a | n | ∅ | $v_6$ |
| m   | ∅ | ∅ | ∅ | a | n | $v_7$ |
| ... |   |   |   |   |   |       |

The loss computation done efficiently by dynamic Programming!

# CTC in recognition of one word



Note: each  $a_i$  is a softmax vector having probabilities of all tokens  
What is the most likely mapping to the text sequence?

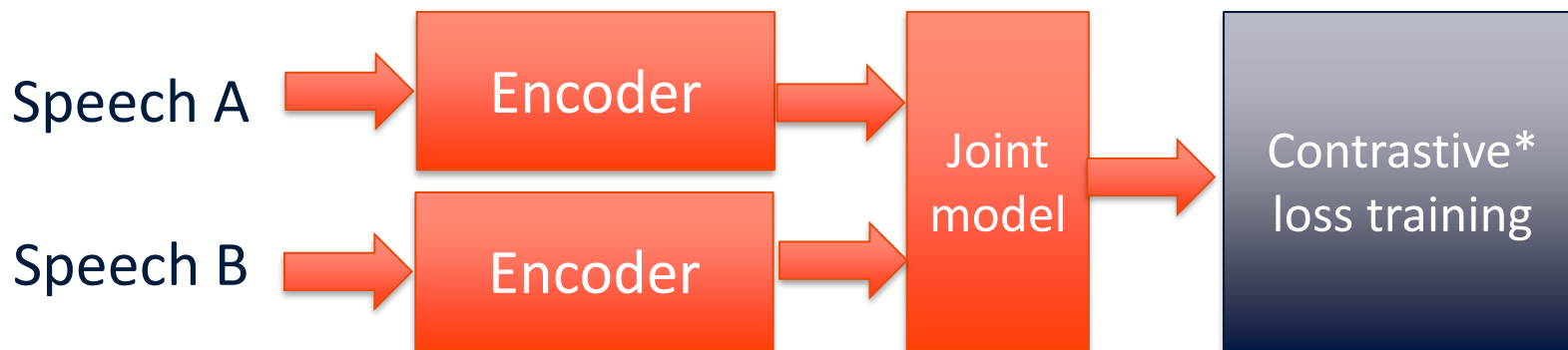


$\emptyset\emptyset man\emptyset$

We can compute a Likelihood for each possible alignment and choose the best.  
But what about other words?

# Speech-to-Speech representations

- “Speech Units”- non-textual embedding of speech sequence for speech seq2seq



# Any Questions on the Lecture



# Now let's form groups of 4



# Discuss Exercises Lecture

# Exercise 1: Speech Signal Representation & Log Mel Filterbank Compression

Raw audio is a continuous acoustic waveform that must be windowed, transformed, and mapped to psychophysically motivated representations prior to neural network processing.

- The lecture outlines the windowing process where speech signals are divided into overlapping frames. Explain why speech signals are non-stationary and why frame lengths of 20–25 ms with a 10 ms step size are used instead of analyzing the entire utterance at once.
- In generating a **Log Mel Spectrogram**, explain why the Mel scale applies linear filter spacing below 1 kHz and logarithmic spacing above 1 kHz. Why are spectral magnitudes converted into a logarithmic Decibel (dB) scale based on Fechner's law?
- Using the data compression calculations (before NN):
  - Given high-quality audio at 48 kHz, down-sampled to wideband 16 kHz (16,000 samples/sec).
  - If you compute a Mel-spectrum using 64 filter bands with a 10 ms hop step (yielding 100 frames/sec), verify the resulting data rate in numbers/second and explain why this preserves only ~13% of the original data.



## Exercise 2: End-to-End ASR Architectures (CTC, RNN-T, and Attention-based Encoder-Decoder)

This lecture introduces three modern end-to-end Automatic Speech Recognition (ASR) paradigms: CTC, RNN-Transducer, and Attention-based Encoder-Decoder (AED).

- Complete a comparison across the three architectures addressing:
  - **Alignment Strategy:** Which rely on explicit frame-level alignment versus implicit cross-attention alignment?
  - **Latency & Streaming:** Which are suited for on-edge streaming applications, and which are typically restricted to offline batch transcription of fixed audio chunks?
  - **Word Error Rate (WER) vs. Complexity:** How do they rank in relative complexity and WER?
- We also introduce **Connectionist Temporal Classification (CTC)**. Explain the functional role of the "blank" symbol  $\emptyset$ . Using the word "*man*", show how multiple valid 11-symbol alignment sequences (e.g., repeating tokens and blanks) map to the same final transcription.
- What architectural design makes **The Conformer** (a "Transformer-CNN sandwich") superior to a pure Transformer for acoustic modeling? What does the CNN component capture compared to the multi-head self-attention module?

## Exercise 3: Classic ASR vs. End-to-End Speech Models & Paralinguistic Representation

Modern speech architectures have evolved from modular statistical systems (GMM-HMM) into unified, multimodal foundation models (Whisper, Meta SeamlessM4T).

- Review the classic ASR pipeline from the 1980s–2010s:
  - What were the distinct roles of the **Acoustic Model** (GMMs/HMMs) and the **Language Model** decoded via Viterbi?
  - How does an end-to-end model like OpenAI's **Whisper** replace this modular pipeline using a single Transformer encoder-decoder?
- What is the standard evaluation metric for ASR accuracy (Word Error Rate - WER)? How is WER computed, and why is transcribing speech to pure text described as a "lossy channel" for human communication?
- Beyond literal text transcription, explain the role of **x-vectors** and self-supervised **Speech Units**. How do these representations enable tasks like speaker verification, emotion detection, and direct "no-text" speech-to-speech translation without generating intermediate text transcripts?



# Discussion



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## On the Text-First Bias: Did NLP Build on the Real Language, or on Its Residue?

Speech is 1.5 million years old; writing is 5,000. Almost everything in this course, from grammars to GPT, was built on the younger one. Writing keeps the words and drops the rest: pitch, pace, hesitation, emphasis, who is speaking and how they feel.

Is text a representation of language, or an artefact that language left behind, which we then mistook for the thing itself?

If most of what a listener receives never reaches the transcript, what does a text-only model understand when it understands a conversation? And what would change in NLP if speech, not text, were the default?

# On Jelinek's Maxim: Can Data Replace the Ear?

“Every time we fire a linguist, the performance of our system goes up” (Jelinek, 1988). For decades speech front-ends were engineered from the biology of hearing: short windows, Mel filters, decibels. Whisper and HuBERT learn from tens of thousands of hours of audio with almost no hand design.

And yet both still start from a log-Mel spectrogram. Did scale make the ear irrelevant, or is the ear the one piece of knowledge that nobody has managed to fire?

What does it mean that our best models of speech begin with a model of the listener? Is that an inductive bias we could drop with more data, or a reminder that speech is defined by being heard?

# On Textless Speech: Are Words Necessary for Meaning?

Speech-to-speech translation used to pass through text: recognise, translate, synthesise. Models like SeamlessM4T now go from waveform to waveform through learned speech units, with no words in between, and carry the tone and the voice across.

When you listen, do you turn sound into words before you understand it, or do words come afterwards, when you try to say what you understood?

If a system can translate meaning and feeling without ever producing a word, are discrete symbols necessary for meaning, or a technology for storing and sharing it? What would be lost in a world that no longer needed the transcript?

# Break



# Questions?



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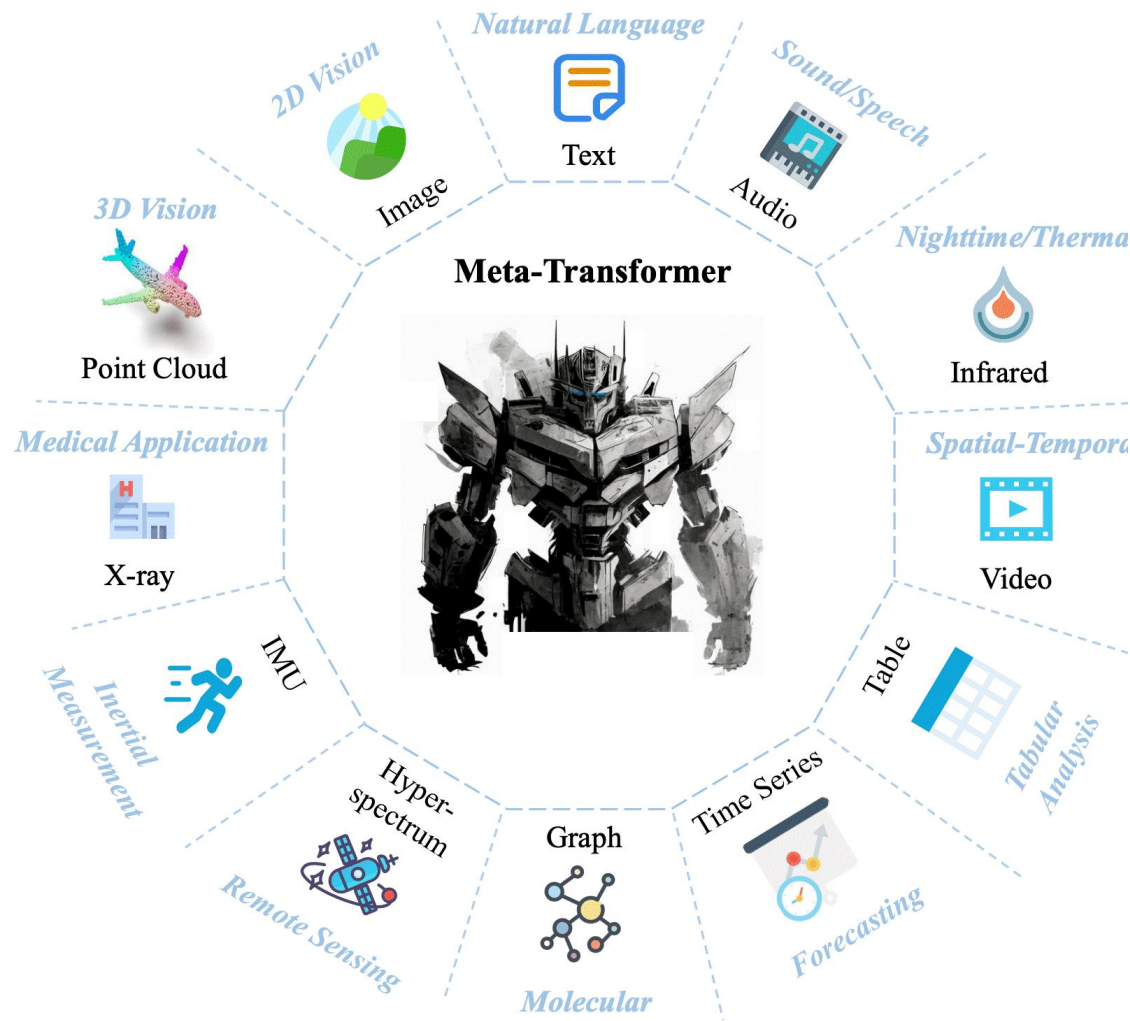
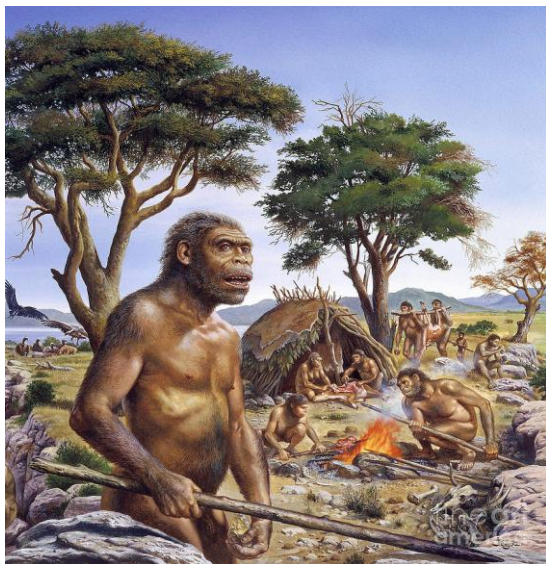
# Multi-Modality



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# When text is not enough

Humans are multimodal agents – unimodal interaction is rare

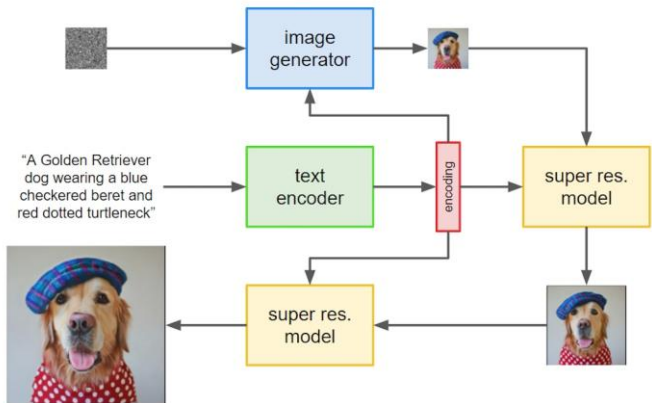




# Text-to-image

Many different types of image generation models exist.

GANs vs Diffusion



Flux-Pro



Flux-Dev



Flux-Schnell



SD3



Dall-E 3



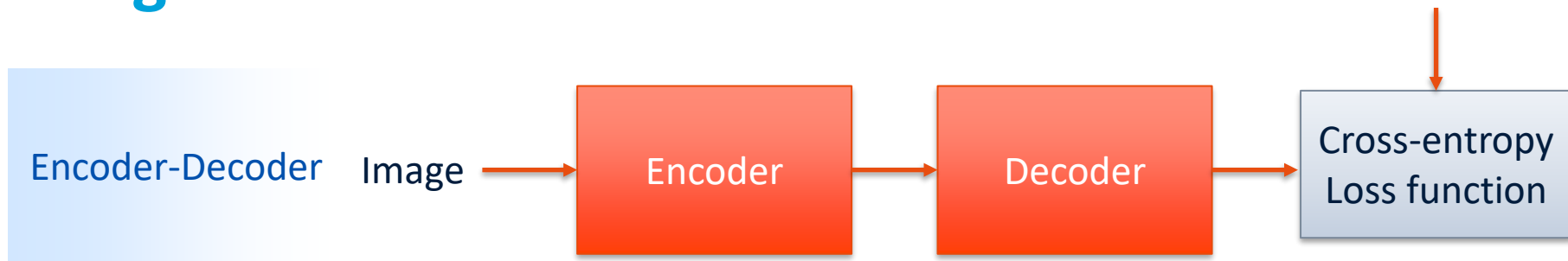
Ideogram



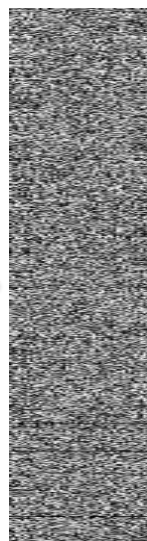
SDXL



# Image-to-Text: encoder-decoder formulations



$3 \times 128 \times 128 = 49152$



“a dog jumping through the snow”

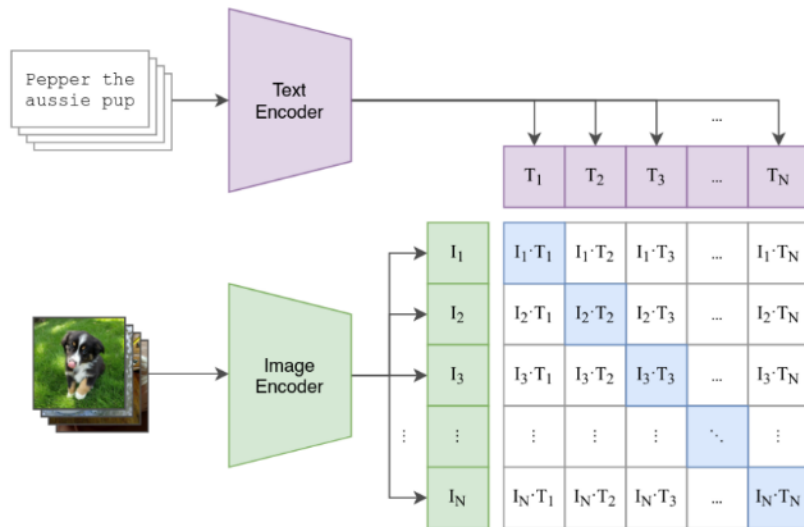
$768 \times 49 = 37632$

# CLIP Image-to-text (Contrastive Learning)

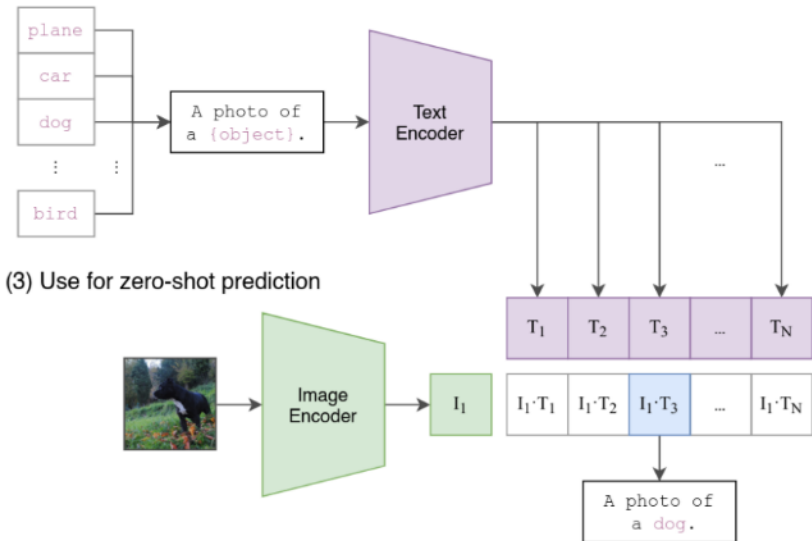
## Learning Transferable Visual Models From Natural Language Supervision

2

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Trained using 400M  
image/text pairs

[\[2103.00020\] Learning Transferable Visual Models From Natural Language Supervision \(arxiv.org\)](#)



# Image-to-text (Contrastive Learning)

## Positive pairs

a white dog in a jump holding a frisbee in his mouth



a woman and child laugh



a large brown dog is crouching in a grassy field



## Negative pairs

a dog running on the sand by the beach



three people with backpacks walk up a road



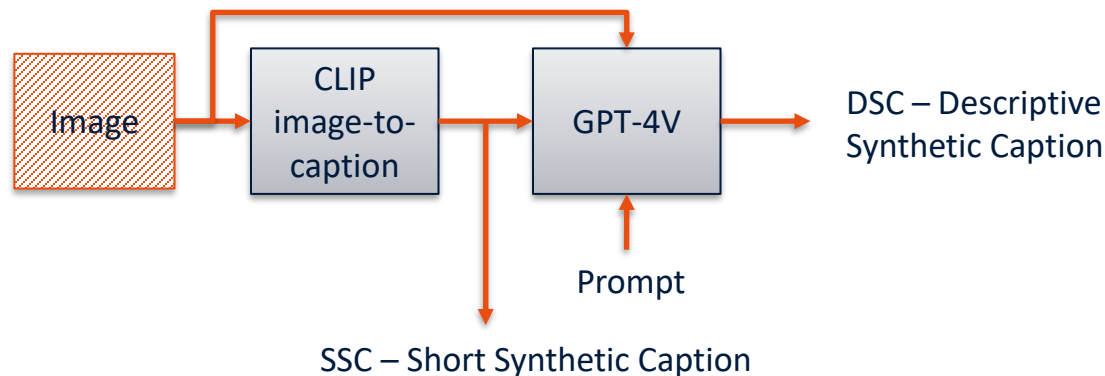
a baby boy vacuuming his house and carpet





# Fusion of synthetic and real data in training

- Open Internet image content has often bad captions (alt-text field)
- “Our data engine, instead of relying on labor-intensive manual annotation, autonomously generates a comprehensive visual dataset called FLD-5B, encompassing a total of 5.4B annotations for 126M images...” (Florence-2, Gemini?)
- DALL-E 3 uses generated image captions in training data to replace original captions: “..a mixture of 95% synthetic captions and 5% ground truth captions”\*



[\\*dall-e-3.pdf \(openai.com\)](#)

# “Chain-of-Frames” – multimodal thinking

- The same as chain-of-thought but for video

Prompt: The background changes to white.

Then:

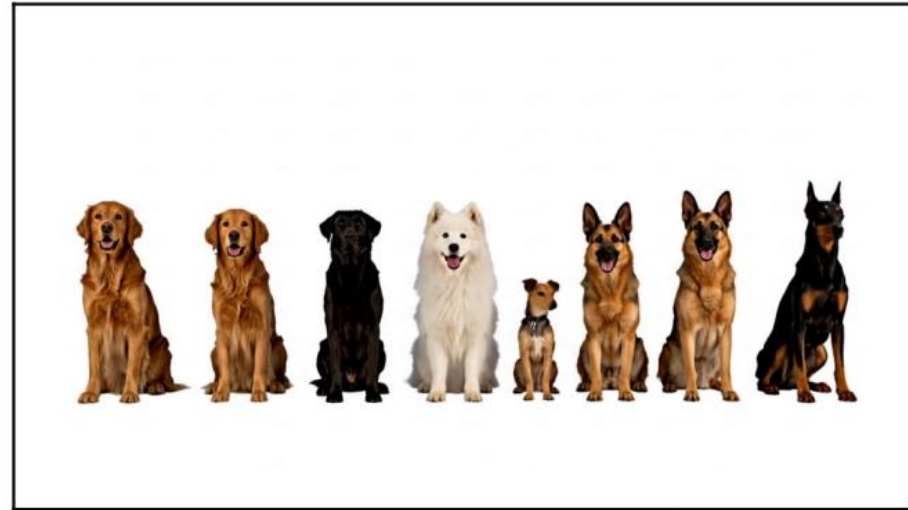
If there is just a single animal: the animal sits in the middle of the image, looking straight at the camera.

If there are multiple animals: all animals line up in a row, with ample white space between them.

Original image



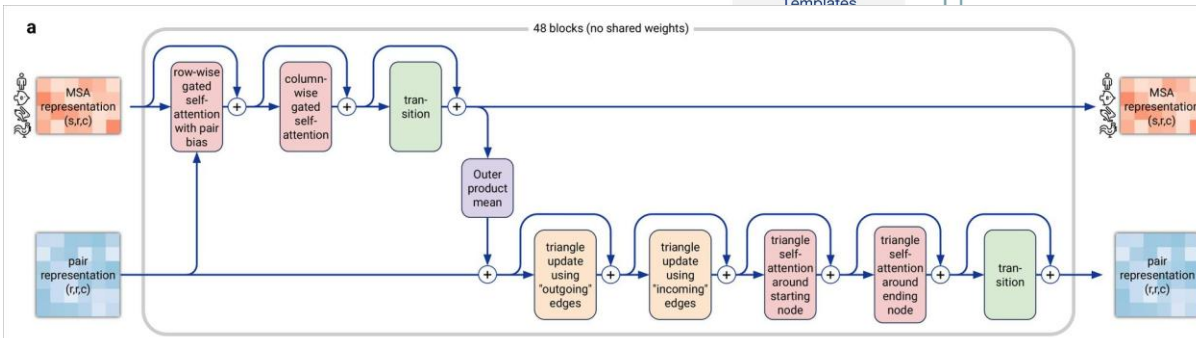
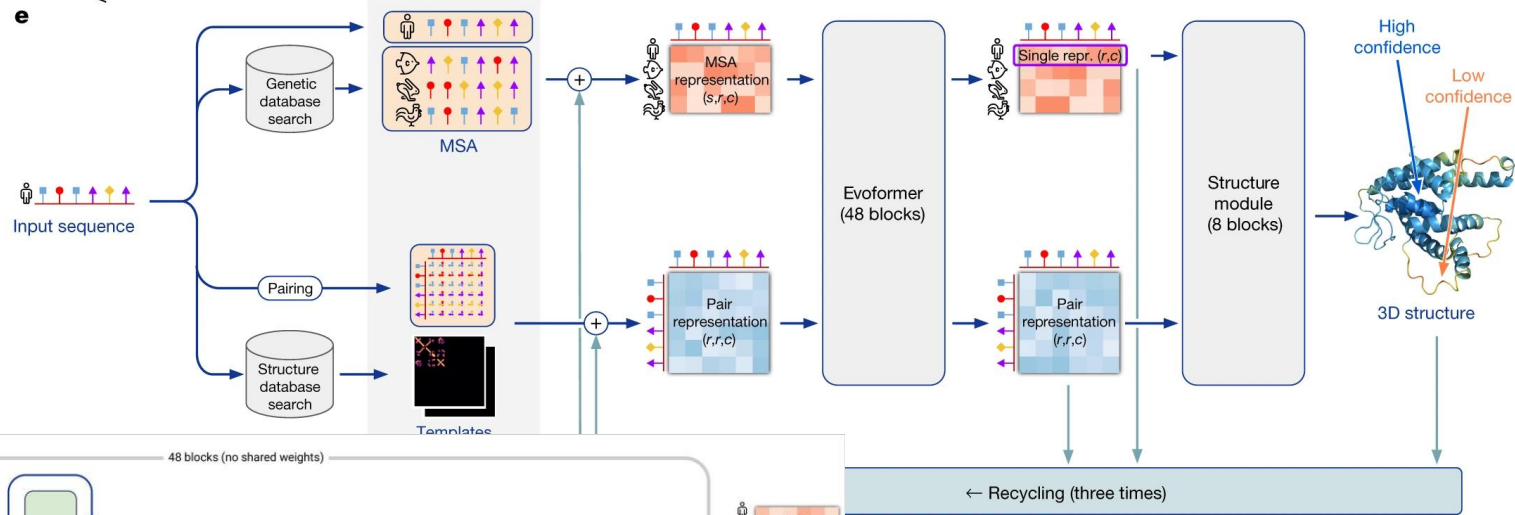
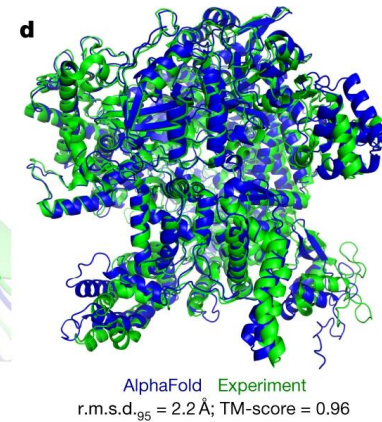
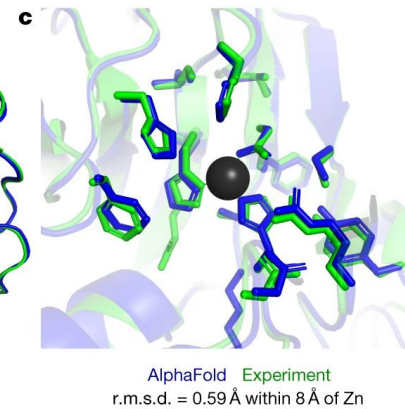
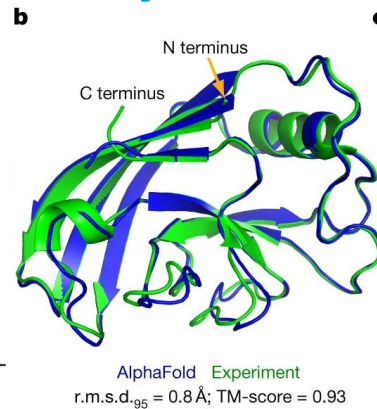
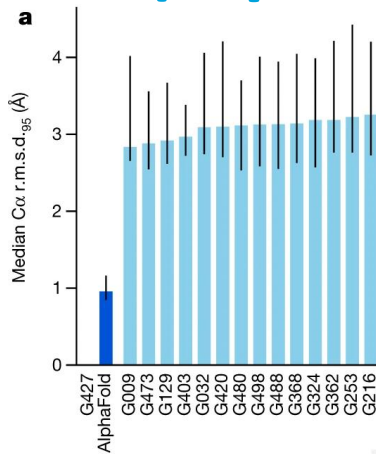
Generated frame (Veo 3)



# Protein data (AlphaFold)

Won Nobel Prize for Chemistry in 2024!

Protein folding:  
linear chain of amino acids to a complex 3D structure



# Any Questions on the Lecture

# Now let's form groups of 4





# Exercise 1: Contrastive Vision-Language Learning and InfoNCE Loss (CLIP)

CLIP (Contrastive Language-Image Pre-training, Radford et al., 2021) jointly trains an image encoder and a text encoder to project both modalities into a shared embedding space using a contrastive objective.

- The lecture presents the symmetric InfoNCE contrastive loss for a matched image-text pair  $(z_i, w_i)$ :

$$\mathcal{L}_i = -\log \frac{\exp(z_i^T w_i / \tau)}{\sum_{j=1}^J \exp(z_i^T w_j / \tau)}$$

Define each term in this equation (including the temperature parameter  $\tau$ , the positive pair dot product  $z_i^T w_i$ , and negative pairs  $z_i^T w_j$ ). What geometric effect does minimizing this loss have on the orientation of image and text vectors in the shared latent space?

- The lectures also illustrates zero-shot classification with CLIP using prompt engineering (e.g., matching image embedding  $w_0$  against prompt texts  $z_0 = \text{"two cats on a couch"}$  and  $z_1 = \text{"a dog playing in a park"}$ ). Explain step-by-step how CLIP performs zero-shot prediction over arbitrary image classes without training a task-specific classification head.
- The lecture demonstrates that the CLIP score can also serve as an automated evaluation metric. How is the cosine similarity between generated images and text captions used to evaluate generative models like DALL-E 3?

## Exercise 2: Multimodal Architecture Formulations (Encoder-Decoder vs. Autoregressive Decoder-Only)

- The lecture slides distinguish between two primary architectural paradigms for Vision-Language Models (VLMs): encoder-decoder cross-attention networks and unified decoder-only autoregressive models.
  - In the **Encoder-Decoder** image captioning formulation, how is the visual input processed into patch/token representations, and how does the text decoder ingest these representations via cross-attention?
  - In the **Decoder-Only** formulation for Visual Question Answering, explain how image patch embeddings and tokenized text question tokens are fused before entering the autoregressive decoder. Why is the entire multimodal input treated as a single concatenated sequence of discrete/continuous tokens?
  - Contrast these pure neural foundation models with the classical **Symbolic Semantic Web / MPEG-7** approaches from early slides in the lecture. What are the trade-offs between symbolic, rule-based knowledge graphs (ontologies, RDF triples) and continuous neural embeddings regarding explainability, maintenance effort, and handling of ambiguous visual scenes?

## Exercise 3: Synthetic Data Recaptioning and Vision-Language-Action (VLA) Grounding

Recent advances in multimodal pre-training and downstream robotics rely heavily on synthetic data engines and reinforcement learning feedback.

- The lecture highlights that raw web-scraped image-text pairs often suffer from low-quality, noisy alt-text. Explain how **DALL-E 3** and **Florence-2** use automated recaptioning engines (e.g., Short Synthetic Captions vs. Descriptive Synthetic Captions generated by models like GPT-4V) to improve text-to-image generation fidelity. Why does training on up to 95% synthetic captions improve prompt adherence rather than causing model collapse?
- It also illustrates **VARP** (Vision-Language-Action models optimized by RL). How can a Vision-Language Model (VLM) act as an automated reward evaluator for robotic trajectory rollouts (e.g., scoring robot arm trajectories for the task "to open drawer")?
- We introduce the concept of "**Chain-of-Frames**" in generative video models like Google DeepMind Veo. How does generating intermediate visual frames parallel the step-by-step reasoning seen in textual Chain-of-Thought (CoT) prompting?

# Discussion



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## On Grounding: Did the Semantic Web Lose to a Dot Product, or Did It Win in Disguise?

The Semantic Web grounded symbols in other symbols: a cat is a mammal is an animal, written down by experts. CLIP grounds a caption in pixels: the vector for “two cats on a couch” points the same way as the vector for a photo of two cats on a couch, learned from 400 million pairs.

The symbol grounding problem asks how a symbol can mean anything if it is only ever defined by other symbols. Does a dot product between a learned text space and a learned image space solve that problem, or is a pixel just another symbol to a network?

If continuous embeddings are a compressed, unreadable version of the knowledge graph the experts tried to write, who lost: the symbolic programme, or only the people who had to maintain it? And what does the model know about cats that no graph could say?



# On the Synthetic Data Paradox: Why Does Multimodality Not Collapse?

Training language models on their own output leads to model collapse. Training a text-to-image model on captions that are 95 percent synthetic makes it better. Both facts are in this course.

What is different? The synthetic captions describe real images. Does a second modality anchor generated text to the world and break the loop, or has the collapse simply been postponed to the images, which will soon be synthetic too?

If grounding in another modality is what keeps synthetic data honest, is there a modality that grounds language itself, and what would a model need in order to be in contact with it?

# On the Universal Transformer: Is Everything a Sequence, or Have We Made It One?

The same architecture folds proteins, steers robot arms, forecasts weather and writes poems. To do so, each of these had to be turned into a sequence of tokens first: an amino acid chain, a trajectory, a grid of pressures, a sentence.

Is “sequence” a property of these things, or of our method? What is lost when a protein, a storm or a hand movement is made into a string of discrete steps so that one hammer can hit it?

Multimodal models claim to join many modalities. Or is there only one modality left, the token, with different front-ends? If so, is that a discovery about information, or an accident of what we know how to build?

# Questions?



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